Antenna Theory Assignment

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April 14, 2026

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Solutions

Question 1: Small Current Loop and Duality Theorem

A small circular loop of radius $a \ll \lambda$ lies in the xy-plane with current I_0. Magnetic dipole moment:

$$m = I_0 \pi a^2$$

(a)

Starting with the vector potential:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu I_0 a}{4\pi} \int_0^{2\pi} \hat{\boldsymbol{\phi}}' \frac{e^{-jkR}}{R} \, d\phi'$$

where

$$R = |\mathbf{r} - \mathbf{r}'| \approx r - a \sin\theta \cos(\phi - \phi')$$

Using the far-field approximation:

$$\frac{e^{-jkR}}{R} \approx \frac{e^{-jkr}}{r} e^{jka \sin\theta \cos(\phi - \phi')}$$

Thus,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu I_0 a}{4\pi r} e^{-jkr} \int_0^{2\pi} \hat{\boldsymbol{\phi}}' e^{jka \sin\theta \cos(\phi - \phi')} d\phi'$$

Using small-loop approximation ($ka \ll 1$):

$$e^{jka \sin\theta \cos(\phi - \phi')} \approx 1 + jka \sin\theta \cos(\phi - \phi')$$

Zeroth-order term vanishes, so:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu I_0 a}{4\pi r} e^{-jkr} \cdot jka \sin\theta \int_0^{2\pi} \hat{\boldsymbol{\phi}}' \cos(\phi - \phi') d\phi'$$

Using:

$$\int_0^{2\pi} \hat{\boldsymbol{\phi}}' \cos(\phi - \phi') d\phi' = \pi \hat{\boldsymbol{\phi}}$$

we get:

$$\mathbf{A} = \frac{j\mu k I_0 \pi a^2 \sin\theta}{4\pi r} e^{-jkr} \hat{\boldsymbol{\phi}}$$

Since $m = I_0 \pi a^2$:

$$\mathbf{A} = \frac{j\mu k m \sin\theta}{4\pi r} e^{-jkr} \hat{\boldsymbol{\phi}}$$

Magnetic Field Using:

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A}$$

In the far-field:

$$H_\theta = -\frac{1}{\mu r} \frac{\partial}{\partial r} (r A_\phi)$$

Substituting:

$$H_\theta = -\frac{1}{\mu r} \frac{\partial}{\partial r} \left(r \cdot \frac{j\mu k m \sin\theta}{4\pi r} e^{-jkr} \right)$$

$$= -\frac{1}{\mu r} \left(\frac{j\mu k m \sin\theta}{4\pi} (-jk) e^{-jkr} \right)$$

$$H_\theta = -\frac{k^2 m \sin\theta}{4\pi r} e^{-jkr}$$

Electric Field Using:

$$\mathbf{E} = \eta\, \mathbf{H} \times \hat{\mathbf{r}}$$

$$E_\phi = \eta H_\theta$$

$$E_\phi = -\eta \frac{k^2 m \sin\theta}{4\pi r} e^{-jkr}$$

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(b)

Given Hertzian dipole fields:

$$E_\theta = \frac{j\eta k I_0 \ell \sin\theta}{4\pi r} e^{-jkr}, \quad H_\phi = \frac{E_\theta}{\eta}$$

Apply duality:

$$\mathbf{E}_A \to \mathbf{H}_F, \quad \mathbf{H}_A \to -\mathbf{E}_F, \quad \eta \to \frac{1}{\eta}, \quad I_0 \ell \to I_m \ell$$

Thus,

$$H_\theta = \frac{jk I_m \ell \sin\theta}{4\pi \eta r} e^{-jkr}$$

$$E_\phi = -\frac{jk I_m \ell \sin\theta}{4\pi r} e^{-jkr}$$

Equivalence Comparing with loop result:

$$\left| \frac{k^2 m \sin\theta}{4\pi r} \right| = \left| \frac{k I_m \ell \sin\theta}{4\pi \eta r} \right|$$

$$km = \frac{I_m \ell}{\eta}$$

$$I_m \ell = \eta k m = \eta k I_0 \pi a^2$$

Dimensional Check

$$[I_m \ell] = [\eta][k][m]$$

$$= \left[\frac{V}{A}\right] \cdot \left[\frac{1}{m}\right] \cdot [A \cdot m^2] = [V \cdot m]$$

Hence, dimensions are consistent.

Question 2: Uniqueness Theorem — The Critical Role of Loss

(a)

Let two field solutions $\{\mathbf{E}_1, \mathbf{H}_1\}$ and $\{\mathbf{E}_2, \mathbf{H}_2\}$ satisfy Maxwell's equations in a bounded volume V with the same sources and identical tangential $\mathbf{E}$ on the boundary surface S.

Define the difference fields:

$$\delta\mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2, \quad \delta\mathbf{H} = \mathbf{H}_1 - \mathbf{H}_2$$

Using the complex Poynting theorem:

$$\int_V \nabla \cdot (\delta\mathbf{E} \times \delta\mathbf{H}^*)\, dV = \oint_S (\delta\mathbf{E} \times \delta\mathbf{H}^*) \cdot d\mathbf{S}$$

Applying the vector identity:

$$\nabla \cdot (\delta\mathbf{E} \times \delta\mathbf{H}^*) = \delta\mathbf{H}^* \cdot (\nabla \times \delta\mathbf{E}) - \delta\mathbf{E} \cdot (\nabla \times \delta\mathbf{H}^*)$$

Thus,

$$\int_V [\delta\mathbf{H}^* \cdot (\nabla \times \delta\mathbf{E}) - \delta\mathbf{E} \cdot (\nabla \times \delta\mathbf{H}^*)]\, dV = \oint_S (\delta\mathbf{E} \times \delta\mathbf{H}^*) \cdot d\mathbf{S}$$

Since tangential $\mathbf{E}_1 = \mathbf{E}_2$ on S, we have:

$$\delta\mathbf{E}_t = 0 \;\Rightarrow\; \delta\mathbf{E} \parallel \hat{\mathbf{n}}$$

Hence,

$$(\delta\mathbf{E} \times \delta\mathbf{H}^*) \cdot \hat{\mathbf{n}} = 0$$

$$\Rightarrow \oint_S (\delta\mathbf{E} \times \delta\mathbf{H}^*) \cdot d\mathbf{S} = 0$$

Therefore,

$$\int_V \nabla \cdot (\delta \mathbf{E} \times \delta \mathbf{H}^*)\, dV = 0$$

Using Maxwell's equations:

$$\nabla \times \delta \mathbf{E} = -j\omega\mu\, \delta \mathbf{H}$$

$$\nabla \times \delta \mathbf{H} = (\sigma + j\omega\epsilon)\delta \mathbf{E}$$

Substituting:

$$\int_V [\delta \mathbf{H}^* \cdot (-j\omega\mu\delta \mathbf{H}) - \delta \mathbf{E} \cdot (\sigma + j\omega\epsilon)^* \delta \mathbf{E}^*]\, dV = 0$$

Taking real part and writing:

$$\epsilon = \epsilon' - j\epsilon'', \quad \mu = \mu' - j\mu''$$

we obtain:

$$-\int_V \left[\sigma|\delta \mathbf{E}|^2 + \omega\epsilon''|\delta \mathbf{E}|^2 + \omega\mu''|\delta \mathbf{H}|^2\right] dV = 0$$

Implication: Since each term is non-negative, the integral can be zero only if:

$$\delta \mathbf{E} = 0, \quad \delta \mathbf{H} = 0$$

Hence, the solution is **unique** in a lossy medium.
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(b)

For a lossless medium:

$$\sigma = 0, \quad \epsilon'' = 0, \quad \mu'' = 0$$

Then the real part becomes:

$$\int_V 0\, dV = 0$$

This does **not** imply:

$$\delta \mathbf{E} = 0, \quad \delta \mathbf{H} = 0$$

conclusion: The key step that fails is the conclusion that the integrand must vanish. In a lossless medium, there are no dissipative terms, so the energy-based argument cannot enforce uniqueness.

Physical Example: Consider a **lossless cavity resonator**. Even with the same boundary conditions and no sources, multiple field distributions (modes) can exist at resonant frequencies.

For example:

- Standing wave patterns inside a perfectly conducting cavity
- Different eigenmodes with the same boundary conditions

Thus, multiple valid solutions exist, demonstrating **non-uniqueness** in a lossless region.

Question 3: LTE (4G) Base Station Coverage

Given:

$$f = 1.8\,\text{GHz}, \quad P_T = 20\,\text{W} = 43\,\text{dBm}$$
$$G_T = 18\,\text{dBi}, \quad G_R = 0\,\text{dBi}$$
$$\eta_T = 0.90, \quad \eta_R = 0.75$$
$$P_{\min} = -100\,\text{dBm}$$

Wavelength:

$$\lambda = \frac{3 \times 10^8}{1.8 \times 10^9} = 0.167\,\text{m}$$

(a)

Using Friis transmission equation:

$$P_R = P_T G_T G_R \eta_T \eta_R \left(\frac{\lambda}{4\pi R}\right)^2$$

At maximum range:

$$P_{\min} = P_T G_T G_R \eta_T \eta_R \left(\frac{\lambda}{4\pi R_{\max}}\right)^2$$

Converting to dB:

$$P_{\min} = P_T + G_T + G_R + 10\log\eta_T + 10\log\eta_R + 20\log\left(\frac{\lambda}{4\pi}\right) - 20\log R_{\max}$$

Substituting:

$$-100 = 43 + 18 + 0 + 10\log(0.9) + 10\log(0.75) + 20\log\left(\frac{0.167}{4\pi}\right) - 20\log R_{\max}$$

$$10\log(0.9) = -0.46\,\text{dB}, \quad 10\log(0.75) = -1.25\,\text{dB}$$

$$20\log\left(\frac{0.167}{4\pi}\right) \approx -37.53\,\text{dB}$$

$$-100 = 43 + 18 - 0.46 - 1.25 - 37.53 - 20\log R_{\max}$$

$$-100 = 21.76 - 20\log R_{\max}$$

$$-20\log R_{\max} = -121.76$$

$$\log R_{\max} = 6.088$$

$$R_{\max} = 10^{6.088} \approx 1.22 \times 10^6\,\text{m}$$

$$R_{\max} \approx 1220\,\text{km}$$

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(b)

Coverage area:

$$A = \pi R_{\text{max}}^2 = \pi(1220)^2$$

$$A \approx 4.68 \times 10^6 \, \text{km}^2$$

Given practical cell radius:

$$R_{\text{real}} = 10 \, \text{km}$$

Overestimation factor:

$$\frac{R_{\text{max}}}{R_{\text{real}}} = \frac{1220}{10} = 122$$

$$\text{Overestimation} \approx 10^2$$

Reason for Overestimation In free space, path loss follows:

$$P_R \propto R^{-2}$$

However, in real-world environments:

$$P_R \propto R^{-n}, \quad n > 2$$

Typical values:

$$n \approx 3 \text{ to } 4$$

Explanation The higher path loss exponent arises due to:

- Multipath propagation (reflection, diffraction, scattering)
- Obstructions (buildings, terrain)

Thus, signals decay much faster than in free space, leading to significantly smaller practical coverage.

Question 6: Quarter-Wave Monopole and Image Theory

(a)

A quarter-wave ($\lambda/4$) vertical monopole is placed on an infinite PEC ground plane.

Using **image theory**, the PEC ground plane can be replaced by an image of the monopole:

- The image is another $\lambda/4$ monopole located below the ground plane
- It carries the same current I_0 and is oriented symmetrically

Thus, the monopole + image together form an equivalent:

$$\text{Half-wave dipole of length } \lambda/2 \text{ in free space}$$

This equivalent problem is valid in the region:

$$z > 0$$

Hence, the far-field of the monopole in the upper half-space is identical to that of a half-wave dipole:

$$E_\theta^{\text{monopole}} = E_\theta^{\text{dipole}}, \quad 0 \le \theta \le \frac{\pi}{2}$$

Using the known result for a center-fed half-wave dipole:

$$E_\theta = \frac{j\eta I_0}{2\pi r} \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} e^{-jkr}$$

Thus,

$$E_\theta^{\text{monopole}} = \frac{j\eta I_0}{2\pi r} \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} e^{-jkr}, \quad 0 \le \theta \le \frac{\pi}{2}$$

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(b)

Given radiation resistance of half-wave dipole:

$$R_{\text{rad}}^{\text{dipole}} = 73\,\Omega$$

Radiation resistance is defined as:

$$P = \frac{1}{2} I_0^2 R_{\text{rad}}$$

Using image theory:

- The monopole radiates only into the upper half-space
- The dipole radiates into the full space
- Radiation pattern in the upper half-space is identical

Thus,

$$P_{\text{mono}} = \frac{1}{2} P_{\text{dipole}}$$

Since:

$$P \propto R_{\text{rad}}$$

we get:

$$R_{\text{rad}}^{\text{mono}} = \frac{1}{2} R_{\text{rad}}^{\text{dipole}}$$

$$R_{\text{rad}}^{\text{mono}} = \frac{73}{2} = 36.5\,\Omega$$

Conclusion:

$$R_{\text{rad}}^{\text{mono}} = 36.5\,\Omega$$

Question 7: Effective Aperture and the Friis Formula

(a)

Starting from the Friis transmission equation (lossless, matched case):

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2$$

The incident power density at the receiving antenna is:

$$S_{\text{inc}} = \frac{P_t G_t}{4\pi R^2}$$

Effective aperture is defined as:

$$A_e \equiv \frac{P_r}{S_{\text{inc}}}$$

Substituting:

$$A_e = \frac{P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2}{\frac{P_t G_t}{4\pi R^2}}$$

$$A_e = G_r \left(\frac{\lambda^2}{(4\pi)^2}\right) \cdot \frac{4\pi R^2}{R^2}$$

$$A_e = \frac{G_r \lambda^2}{4\pi}$$

Result:

$$A_e = \frac{G_r \lambda^2}{4\pi}$$

Thus, the **gain and aperture correspond to the receiving antenna**.
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(b)

Given:

$$D_{\max} = 1.64, \quad \eta_{\text{rad}} = 1$$

$$G = \eta_{\text{rad}} D_{\max} = 1.64$$

$$f = 500\,\text{MHz}, \quad S_{\text{inc}} = 10\,\mu\text{W/m}^2 = 10 \times 10^{-6}\,\text{W/m}^2$$

Wavelength:

$$\lambda = \frac{3 \times 10^8}{500 \times 10^6} = 0.6\,\text{m}$$

(i) Effective Aperture

$$A_e = \frac{G\lambda^2}{4\pi} = \frac{1.64 \times (0.6)^2}{4\pi}$$

$$A_e = \frac{1.64 \times 0.36}{12.566} \approx 0.04698\,\mathrm{m}^2$$

$$A_e \approx 470\,\mathrm{cm}^2$$

(ii) Received Power

$$P_r = A_e S_{\text{inc}}$$

$$P_r = 0.04698 \times 10 \times 10^{-6}$$

$$P_r = 4.7 \times 10^{-7}\,\mathrm{W}$$

$$P_r = 0.47\,\mu\mathrm{W}$$

Final Answers:

$$A_e \approx 470\,\mathrm{cm}^2, \quad P_r \approx 0.47\,\mu\mathrm{W}$$

Question 9

(b) Near-field / Far-field Transition

The transition point between the near-field and far-field regions is obtained by equating the magnitudes of the two field components:

$$\frac{k}{r} = \frac{1}{r^2}.$$

Multiplying both sides by r^2 gives:

$$kr = 1.$$

Thus, the transition occurs when $kr = 1$.

To obtain a physical distance, we fix the frequency at $f = 1$ GHz. The corresponding wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^9} = 0.3 \text{ m},$$

and the wavenumber is

$$k = \frac{2\pi}{\lambda}.$$

Using the transition condition $kr = 1$, we get:

$$r = \frac{1}{k} = \frac{\lambda}{2\pi}.$$

Substituting $\lambda = 0.3$ m,

$$r = \frac{0.3}{2\pi} \approx 0.048 \text{ m}.$$

Expressed in terms of wavelength,

$$r \approx 0.16\lambda.$$

Interpretation: This result shows that the near-field region dominates very close to the antenna, where fields decay rapidly as $1/r^2$. Beyond the transition point ($kr > 1$), the far-field term dominates and the field behaves as a radiating wave decreasing as $1/r$. Thus, the transition marks the distance beyond which true radiation behavior is observed.

Question 10: Friis Radar Range Equation

Given radar range equation:

$$P_R = \frac{P_T G_T G_R \lambda^2 \sigma}{(4\pi)^3 R^4}$$

Given:

$$f = 10\,\text{GHz}, \quad \lambda = \frac{3 \times 10^8}{10 \times 10^9} = 0.03\,\text{m}$$

$$P_T = 50\,\text{kW} = 50 \times 10^3\,\text{W}$$

$$G_T = G_R = 30\,\text{dBi}, \quad P_{\min} = -90\,\text{dBm}$$

Convert P_T to dBm:

$$P_T = 50 \times 10^3 = 5 \times 10^4\,\text{W}$$

$$P_T(\text{dBm}) = 10\log\big(5 \times 10^4 \times 10^3\big) = 10\log\big(5 \times 10^7\big) \approx 76.99\,\text{dBm}$$

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(a) $\sigma = 1\,\mathbf{m}^2$

Radar equation in dB:

$$P_R = P_T + G_T + G_R + 20\log\lambda + 10\log\sigma - 30\log(4\pi) - 40\log R$$

Substitute values:

$$-90 = 76.99 + 30 + 30 + 20\log(0.03) + 0 - 30\log(4\pi) - 40\log R$$

$$20\log(0.03) = -30.46\,\text{dB}, \quad 30\log(4\pi) \approx 32.98\,\text{dB}$$

$$R_{\max} \approx 2.18\,\text{km}$$

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(b) $\sigma = 10^{-3}\,\mathbf{m}^2$

$$10\log\sigma = 10\log\big(10^{-3}\big) = -30\,\text{dB}$$

$$R_{\max} \approx 388.057\,\text{m}$$

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Range Reduction Factor

$$\frac{R_2}{R_1} = \frac{388.057}{2180} \approx 0.177$$

$$\text{Reduction (dB)} = 10 \log(0.177) \approx -7.495\,\text{dB}$$

Final Results:

$$R_{\sigma=1} \approx 2.18\,\text{km}, \quad R_{\sigma=10^{-3}} \approx 388.057\,\text{m}$$

$$\text{Range reduction factor} \approx 0.177\ (\approx -7.495\,\text{dB})$$